Boss Challenge — Paper 32 (Class 7)

Soil | Wastewater Story | Comparing Quantities | Algebraic Expressions

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

---

1. The slow breaking down of rocks into tiny particles, over very long times, to form soil is called:

(A) melting
(B) weathering
(C) flooding
(D) digging

2. Water that has been used by homes, factories and farms, and so made dirty, is called:

(A) groundwater
(B) wastewater (sewage)
(C) rainwater
(D) distilled water

3. A ratio is a way of comparing two quantities of the same kind by:

(A) addition
(B) subtraction
(C) division (how many times one is of the other)
(D) rounding off

4. A symbol such as x, which can stand for many different number values, is called a:

(A) variable
(B) constant
(C) equation
(D) product

5. The dark, crumbly material in soil, made from the decayed remains of plants and animals, is called:

(A) gravel
(B) sand
(C) clay
(D) humus

6. The network of underground pipes that carries sewage away from our homes is called the:

(A) sewerage
(B) canal
(C) pipeline of drinking water
(D) electric cable

7. The ratio 4 : 6 written in its simplest form is:

(A) 8 : 12
(B) 2 : 3
(C) 4 : 6
(D) 3 : 2

8. A fixed number such as 7, whose value never changes, is called a:

(A) coefficient
(B) variable
(C) term
(D) constant

9. The set of distinct layers you see when you cut down through the ground is called the:

(A) crop pattern
(B) soil profile
(C) water table
(D) food chain

10. The special place where dirty water is cleaned before being let out is called a:

(A) water tank
(B) wastewater treatment plant
(C) pumping station
(D) power plant

11. To change the fraction 3/4 into a percentage you multiply it by 100, which gives:

(A) 300%
(B) 34%
(C) 75%
(D) 43%

12. In the term 5x, the number 5 that multiplies the variable x is called the:

(A) variable
(B) constant
(C) exponent
(D) coefficient

13. The topmost, dark layer of soil, richest in humus, where most plants spread their roots, is the:

(A) bedrock
(B) subsoil
(C) water table
(D) topsoil

14. At a treatment plant, large floating objects like rags, sticks and plastic are first removed by:

(A) a settling tank
(B) an aerator
(C) bar screens
(D) a chimney

15. The percentage 25% written as a fraction in its simplest form is:

(A) 25/10
(B) 1/2
(C) 1/4
(D) 1/25

16. The parts of an expression that are separated by + or - signs are called its:

(A) terms
(B) powers
(C) factors
(D) coefficients

17. Among the three particle types - sand, silt and clay - the LARGEST particles are:

(A) sand
(B) clay
(C) silt
(D) humus

18. The tank in which heavy sand, grit and small pebbles settle out of the sewage is the:

(A) grit and sand removal tank
(B) aeration tank
(C) biogas tank
(D) overhead tank

19. Expressed as a percentage, the decimal 0.6 becomes:

(A) 0.6%
(B) 60%
(C) 6%
(D) 600%

20. Terms that have exactly the same variable factors, such as 3x and 7x, are called:

(A) like terms
(B) coefficients
(C) unlike terms
(D) constants

21. Among sand, silt and clay, the SMALLEST particles are:

(A) silt
(B) gravel
(C) sand
(D) clay

22. In a settling tank, the solid waste that sinks and collects at the bottom is called:

(A) sludge
(B) scum
(C) clear water
(D) biogas

23. 20% of 150 is:

(A) 20
(B) 30
(C) 3
(D) 300

24. The terms 3x and 3y are:

(A) like terms
(B) equal terms
(C) constant terms
(D) unlike terms

25. A soil through which water passes very quickly and which dries out fast is mostly:

(A) loamy soil
(B) sandy soil
(C) clayey soil
(D) frozen soil

26. In a settling tank, the oil, grease and other light matter that float on top are skimmed off as:

(A) scum
(B) sludge
(C) clean water
(D) manure

27. If 2 pens cost ■30, then by the unitary method 5 pens cost:

(A) ■150
(B) ■60
(C) ■75
(D) ■15

28. An expression having exactly ONE term, such as 4xy, is called a:

(A) monomial
(B) binomial
(C) trinomial
(D) equation

29. A soil that holds a great deal of water and feels sticky when wet is mostly:

(A) clayey soil
(B) loamy soil
(C) rocky soil
(D) sandy soil

30. Air is bubbled through the cleared water in the aeration tank mainly to help:

(A) make the water look fizzy for sale
(B) useful aerobic bacteria grow and eat the wastes
(C) add colour to the water
(D) freeze the water solid

31. When the selling price of an article is MORE than its cost price, the seller makes a:

(A) discount
(B) profit
(C) interest
(D) loss

32. An expression having exactly TWO terms, such as x + 5, is called a:

(A) trinomial
(B) binomial
(C) constant
(D) monomial

33. The soil that is best for growing most crops, being a balanced mixture of sand, silt, clay and humus, is:

(A) loamy soil
(B) pure sandy soil
(C) pure clayey soil
(D) gravelly soil

34. The bacteria that break down the wastes in the aeration tank are able to work only when there is:

(A) salt water
(B) no air at all
(C) oxygen (air)
(D) bright sunlight

35. A toy is bought for ■200 and sold for ■250. The profit is:

(A) ■200
(B) ■450
(C) ■50
(D) ■250

36. An expression having exactly THREE terms is called a:

(A) binomial
(B) monomial
(C) variable
(D) trinomial

37. The downward movement of water through soil, measured as the amount that sinks in a given time, is the:

(A) percolation rate
(B) evaporation rate
(C) germination rate
(D) erosion rate

38. The cleaned water leaving the treatment plant is usually:

(A) burned in a furnace
(B) sent straight back to kitchen taps
(C) released into a river or sea, or used to water gardens
(D) buried deep underground forever

39. A toy costing ■200 is sold for ■250, a profit of ■50. The profit per cent is:

(A) 20%
(B) 5%
(C) 50%
(D) 25%

40. Adding the like terms 6a + 2a gives:

(A) 8a
(B) 8
(C) 12a
(D) $8a^2$

41. Sandy soil has a HIGH percolation rate mainly because:

(A) it is always found near rivers
(B) its large particles leave big air gaps for water to pass
(C) it contains a lot of humus
(D) it has tiny tightly-packed particles

42. The settled sludge is moved to a closed tank where bacteria break it down and produce a useful fuel gas called:

(A) steam
(B) oxygen
(C) biogas
(D) petrol

43. A shirt costing ■400 is sold for ■360. The loss per cent is:

(A) 10%
(B) 40%
(C) 90%
(D) 4%

44. Simplifying 9y - 4y gives:

(A) 5y
(B) 13y
(C) 5
(D) $5y^2$

45. Clayey soil is well suited to a crop like paddy (rice) because clay:

(A) never needs any rain
(B) contains no minerals at all
(C) lets water drain away at once
(D) holds water for a long time

46. Untreated sewage flowing through open drains is dangerous mainly because it:

(A) makes the water taste sweet
(B) cools the whole neighbourhood
(C) spreads diseases like cholera and typhoid
(D) turns into clean drinking water

47. Two ratios are said to be in proportion when:

(A) the two ratios are equal
(B) one is double the other
(C) both contain the number 1
(D) their sums are equal

48. Written in algebra, 'a number x made 5 greater' is the expression:

(A) x / 5
(B) 5x
(C) x + 5
(D) x - 5

49. The hard, solid, unbroken rock found at the very bottom of a soil profile is the:

(A) subsoil
(B) humus
(C) topsoil
(D) bedrock

50. Dissolved substances in sewage such as nitrogen and phosphorus compounds are examples of its:

(A) harmless food colours
(B) pure oxygen gas
(C) clean minerals safe to drink
(D) contaminants (nutrients)

51. In the proportion 3 : 5 = 9 : ?, the missing number is:

(A) 11
(B) 45
(C) 15
(D) 7

52. The algebraic expression for 'twice a number n' is:

(A) n / 2
(B) 2n
(C) n^2
(D) n + 2

53. Earthworms are called the farmer's friend mainly because they:

- **(A)** make burrows that let air and water into the soil
- **(B)** stop all rain from falling
- **(C)** eat all the crop seeds
- **(D)** turn clay into pure sand

54. Keeping our surroundings clean and disposing of human and other wastes properly is called:

- **(A)** decoration
- **(B)** irrigation
- **(C)** ventilation
- **(D)** sanitation

55. The extra money paid for the use of borrowed money over time is called:

- **(A)** simple interest
- **(B)** profit
- **(C)** principal
- **(D)** discount

56. The value of the expression $2x + 3$ when $x = 4$ is:

- **(A)** 9
- **(B)** 11
- **(C)** 14
- **(D)** 24

57. The carrying away of fertile topsoil by wind or running water is called:

- **(A)** weathering
- **(B)** percolation
- **(C)** irrigation
- **(D)** soil erosion

58. Used cooking oil should NOT be poured down the kitchen drain because it:

- **(A)** kills no germs and helps flow
- **(B)** makes the water cleaner
- **(C)** turns into safe drinking water
- **(D)** hardens and blocks the drain pipes

59. The formula for simple interest, with principal P, rate R% per year and time T years, is:

- **(A)** $SI = P \times R \times T \times 100$
- **(B)** $SI = P + R + T$
- **(C)** $SI = (P \times R \times T) / 100$
- **(D)** $SI = P / (R \times T)$

60. The value of x^2 when $x = 5$ is:

- **(A)** 25
- **(B)** 7
- **(C)** 10
- **(D)** 52

61. Planting trees and grass on a bare hillside helps mainly by:

(A) making the soil into solid rock
(B) removing all the humus
(C) holding the soil with roots and preventing erosion
(D) stopping the sun from ever shining

62. A low-cost toilet that turns human waste into useful manure with the help of worms is a:

(A) rainwater tank
(B) flush-and-forget sewer toilet
(C) vermi-processing toilet
(D) solar water heater

63. For ■1000 lent at 5% per year over a period of 2 years, the simple interest works out to:

(A) ■50
(B) ■1000
(C) ■100
(D) ■200

64. Adding the expressions $3x + 2$ and $5x + 4$ gives:

(A) $8x^2 + 6$
(B) $8x + 6$
(C) $8x + 8$
(D) $15x + 6$

65. The amount of water a soil can keep after the extra has drained away is called its:

(A) percolation rate
(B) water-holding capacity
(C) particle size
(D) boiling point

66. Paints, medicines and other chemicals must not be poured into drains because they:

(A) add useful vitamins to the water
(B) kill the helpful bacteria that clean the sewage
(C) turn the sewage into pure water
(D) make the bacteria grow faster

67. The total money to be repaid on a loan, called the amount, is equal to:

(A) principal x interest
(B) interest only
(C) principal + interest
(D) principal - interest

68. Subtracting $2a$ from $7a$ gives:

(A) $2a - 7a$
(B) 5
(C) $9a$
(D) $5a$

69. Clayey soil has a LOW percolation rate because its tiny particles:

(A) pack tightly together, leaving very little space for water
(B) float away in the wind easily
(C) are very large with wide gaps
(D) are made of pure humus

70. After it is dried, the treated sludge can be put to good use as:

(A) window glass
(B) drinking water
(C) manure for fields
(D) cooking fuel oil

71. In a class of 40 students, 10 are absent. The percentage of students absent is:

(A) 10%
(B) 25%
(C) 75%
(D) 40%

72. In the term -3pq, the variable factors are:

(A) 3 and q
(B) p and q
(C) -3 and p
(D) p, q and 3

73. Mixing dead leaves and cow dung into a field improves the soil mainly by:

(A) removing all its water
(B) making it unfit for any crop
(C) turning the soil into clay
(D) increasing its humus and making it more fertile

74. Where there is no underground sewer, household sewage can be collected and partly treated on the spot in a:

(A) septic tank
(B) biogas cylinder
(C) petrol tank
(D) overhead water tank

75. To find what per cent 15 is of 60, you should compute:

(A) 15 + 60
(B) (60 / 15) x 100
(C) 15 x 60
(D) (15 / 60) x 100

76. The perimeter of a square whose side is s units is written as the expression:

(A) s + 4
(B) s■
(C) 2s
(D) 4s

77. Arranged from LARGEST to smallest, the three soil particle types are:

(A) clay, sand, silt
(B) clay, silt, sand
(C) sand, silt, clay
(D) silt, sand, clay

78. Diseases such as typhoid, cholera and dysentery spread mainly through:

(A) dry sand
(B) bright sunlight
(C) clean filtered air
(D) water contaminated by sewage

79. Increasing 50 by 10% gives:

(A) 60
(B) 45
(C) 500
(D) 55

80. The algebraic expression for 'a number y decreased by 8' is:

(A) 8 - y
(B) 8y
(C) y + 8
(D) y - 8

81. The mixing of harmful materials like plastic and chemicals into the soil is called:

(A) percolation
(B) soil pollution
(C) weathering
(D) soil erosion

82. Throwing tea leaves, solid food scraps and cotton into the kitchen sink is wrong because they:

(A) make the water drinkable
(B) clog the drains
(C) clean the pipes from inside
(D) turn into safe gas

83. Decreasing 80 by 25% gives:

(A) 55
(B) 20
(C) 100
(D) 60

84. The number of terms in the expression 2x + 3y - 7 is:

- **(A)** one
- **(B)** three
- **(C)** two
- **(D)** four

85. Sandy-loam soil suits cotton well because it:

- **(A)** never lets any water in
- **(B)** drains well and warms up quickly
- **(C)** stays flooded all year
- **(D)** contains no air at all

86. The first group of steps at a plant that removes solids by screening and letting them settle is called the:

- **(A)** the biogas stage
- **(B)** physical (mechanical) cleaning
- **(C)** the chlorination stage
- **(D)** the bottling stage

87. The ratio of 50 paise to 1 rupee (100 paise), in simplest form, is:

- **(A)** 1 : 2
- **(B)** 1 : 50
- **(C)** 50 : 1
- **(D)** 2 : 1

88. The coefficient of x in the term -x is:

- **(A)** x
- **(B)** 1
- **(C)** -1
- **(D)** 0

89. The layer lying just below the topsoil, harder and with much less humus, is the:

- **(A)** subsoil
- **(B)** topsoil
- **(C)** bedrock
- **(D)** humus

90. An open drain running close to a house is a health risk because it is a:

- **(A)** good place to store food
- **(B)** safe playground for children
- **(C)** breeding place for disease-carrying flies and mosquitoes
- **(D)** source of clean drinking water

91. On a map drawn to a scale of 1 : 1000, a length of 1 cm on the map stands for:

- **(A)** 1000 cm on the ground
- **(B)** 1000 km on the ground
- **(C)** 10 cm on the ground
- **(D)** 1 cm on the ground

92. Soil scientists write the percolation rate r as the water w divided by the time t. As a formula this is:

- **(A)** r = t / w
- **(B)** r = w x t
- **(C)** r = w + t
- **(D)** r = w / t

93. In a percolation test, 200 mL of water sinks through a soil sample in 20 minutes. The percolation rate is:

- **(A)** 4000 mL per minute
- **(B)** 100 mL per minute
- **(C)** 10 mL per minute
- **(D)** 20 mL per minute

94. A town sends 5000 litres of sewage to the plant, and 4000 litres come out as treated water. The percentage of water recovered is:

- **(A)** 20%
- **(B)** 125%
- **(C)** 80%
- **(D)** 40%

95. In a 500 g sample of soil, 100 g is found to be clay. The percentage of clay in the soil is:

- **(A)** 20%
- **(B)** 100%
- **(C)** 50%
- **(D)** 5%

96. A tank already holds 200 litres and gains 100 litres every hour. After h hours it holds:

- **(A)** (200 - 100h) litres
- **(B)** (300h) litres
- **(C)** (200 + 100h) litres
- **(D)** (200 x 100h) litres

97. When wet clay can be rolled into a thin thread without breaking, it shows that clay particles:

- **(A)** cannot hold any water
- **(B)** are the largest of all soils
- **(C)** are very far apart
- **(D)** stick closely together

98. We should use water carefully and avoid wasting it so that:

- **(A)** more germs grow in the pipes
- **(B)** treatment plants run out of work for good
- **(C)** less sewage is produced and less needs to be treated
- **(D)** the river becomes dirtier

99. In a tank, the water is mixed as 3 parts clean water to 1 part sludge. The fraction of the mixture that is sludge is:

- **(A)** 1/3
- **(B)** 1/4
- **(C)** 1/2
- **(D)** 3/4

100. The value of the expression $3n - 1$ when $n = 10$ is:

- **(A)** 31
- **(B)** 2
- **(C)** 27
- **(D)** 29